

Some Properties of Prime-Cubed Cyclotomic Cosets in the Cyclic Group $\mathbb{Z}_m$

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Abstract

Cyclotomic cosets in cyclic and abelian groups have been of interest and extensively studied due to their rich algebraic structures and wide applications. In this project, we focus on q -cyclotomic cosets of the cyclic group $\mathbb{Z}_m$ in the case where $q = p^3$ is a prime-cubed integer and m is a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$, the q -cyclotomic coset in $\mathbb{Z}_m$ containing a is defined to be $S_q(a) = \{q^i \cdot a \mid i = 0, 1, 2, \dots\}$. The q -cyclotomic coset $S_q(a)$ is said to be of type I if $S_q(a) = S_q(-a)$, type II if $S_q(a) = S_q(-pa)$ and type III if $S_q(a) = S_q(-p^2a)$. Properties, characterizations and enumerations of such q -cyclotomic cosets are given. Some number theoretical results can be derived using properties of these cyclotomic cosets.

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Chapter 1

Introduction

Cyclotomic cosets in cyclic and abelian groups have been of interest and extensively studied due to their rich algebraic structures and wide applications. For example, they have a direct link with self-dual cyclic codes and their generalizations. In [1], self-dual cyclic codes of length n over $\mathbb{F}_q$ have been studied based on q -cyclotomic cosets in $\mathbb{Z}_n$ when n is odd and $q = 2^m$ for some positive integer m . Later, Hermitian self-dual quasi-abelian codes are studied in [2] and [4]. Characterization and enumeration of Euclidean self-dual and Euclidean self-orthogonal abelian codes are studied in [3]. They are all related to q -cyclotomic class of some types. In [6], formulas for the number of 2^l -cyclotomic and 2^{2l} -cyclotomic cosets of some types in the cyclic group $\mathbb{Z}_{p^r}$ are simplified.

From explanations above, some types of q -cyclotomic cosets where $q = 2^m$ have been long considered. Therefore, it is interesting to study q -cyclotomic of other types where $q = p^m$ for any prime p and a positive integer m . In this project, we consider $m = 3$ and introduce some types of q -cyclotomic cosets. Precisely, q is a **prime-cubed**. Properties, characterizations and enumerations of such q -cyclotomic cosets will be studied. Some number theoretical results can be derived using properties of these cyclotomic cosets.

Chapter 2

Preliminaries

In this chapter, cyclotomic cosets in the cyclic group $\mathbb{Z}_m$ are recalled. Known properties cyclotomic cosets in $\mathbb{Z}_m$ are discussed together with characterization and enumeration of some subclasses of cyclotomic cosets.

Definition 2.1. For positive integers i and j with $\gcd(i, j) = 1$, the **multiplicative order of j modulo i** , denoted by $\text{ord}_i(j)$, is defined to be the smallest positive integer s such that $j^s \equiv 1 \pmod{i}$, or equivalently, $i \mid j^s - 1$.

Example 2.2. The multiplicative order of 3 modulo 2 is $\text{ord}_2(3) = 1$ because $s = 1$ is the smallest positive integer such that $2 \mid 3^s - 1$. The multiplicative order of 7 modulo 5 is $\text{ord}_5(7) = 4$ because $s = 4$ is the smallest positive integer such that $5 \mid 7^s - 1$. Note that $7^4 = 2401$.

Definition 2.3. For $a \in \mathbb{Z}_m$, the **additive order** of a , denoted by $\text{ord}(a)$, is defined to be the smallest positive integer s such that $s \cdot a = \underbrace{a + a + \cdots + a}_{s \text{ copies}} = 0$.

Example 2.4. In $\mathbb{Z}_{45}$, $\text{ord}(3) = 15$ because $s = 15$ is the smallest positive integer such that $s \cdot 3 = 0$ and $\text{ord}(2) = 45$ because $s = 45$ is the smallest positive integer such that $s \cdot 2 = 0$.

Definition 2.5. For each $a \in \mathbb{Z}_m$ and prime power q such that $\gcd(q, m) = 1$, the **q -cyclotomic coset of $\mathbb{Z}_m$ containing a** is defined to be

$$S_q(a) = \{q^i \cdot a \mid i = 0, 1, 2, \dots\},$$

where $q^i \cdot a := \sum_{j=1}^{q^i} a$ in $\mathbb{Z}_m$. For convenience, denote by $q^i a$ the element $q^i \cdot a \in \mathbb{Z}_m$.

Example 2.6. The 8-cyclotomic coset of $\mathbb{Z}_{45}$ containing 0 is the set $S_8(0) = \{0\}$, containing 1 is the set $S_8(1) = \{1, 8, 17, 19\}$ and containing 5 is the set $S_8(5) = \{5, 40\}$.

Theorem 2.7 ([1, 2, 4]). The q -cyclotomic cosets of $\mathbb{Z}_m$ form a partition of $\mathbb{Z}_m$.

Lemma 2.8 ([5]). Let G be a finite group with the cardinality m . Then $\text{ord}(a) \mid m$ for all $a \in G$.

Lemma 2.9 ([6]). Let q be a prime power and let m be a positive integer such that $\gcd(q, m) = 1$. Then $S_q(a) = \{q^i \cdot a \mid 0 \leq i < \text{ord}_{\text{ord}(a)}(q)\}$ and $|S_q(a)| = \text{ord}_{\text{ord}(a)}(q)$ for all $a \in \mathbb{Z}_m$.

Definition 2.10. Let $a \in \mathbb{Z}_m$ and let q be a prime power. A q -cyclotomic coset $S_q(a)$ is said to be of

Type I if $S_q(a) = S_q(-a)$, or Type I' if $S_q(a) \neq S_q(-a)$.

In addition, if $q = p^3$ is a prime-cubed number for some prime p , a q -cyclotomic coset $S_q(a)$ is said to be of

Type II if $S_q(a) = S_q(-pa)$, Type II' if $S_q(a) \neq S_q(-pa)$,

Type III if $S_q(a) = S_q(-p^2a)$, or Type III' if $S_q(a) \neq S_q(-p^2a)$.

Example 2.11. Let $q = 2^3 = 8$. Then the q -cyclotomic coset in $\mathbb{Z}_{45}$ are

$$\begin{aligned} S_8(0) &= \{0\}, \\ S_8(1) &= S_8(8) = S_8(17) = S_8(19) = \{1, 8, 17, 19\}, \\ S_8(2) &= S_8(16) = S_8(34) = S_8(38) = \{2, 16, 34, 38\}, \\ S_8(3) &= S_8(6) = S_8(12) = S_8(24) = \{3, 6, 12, 24\}, \\ S_8(4) &= S_8(23) = S_8(31) = S_8(32) = \{4, 23, 31, 32\}, \\ S_8(5) &= S_8(40) = \{5, 40\}, \\ S_8(7) &= S_8(11) = S_8(29) = S_8(43) = \{7, 11, 29, 43\}, \\ S_8(9) &= S_8(18) = S_8(27) = S_8(36) = \{9, 18, 27, 36\}, \end{aligned}$$

$$\begin{aligned}
S_8(10) &= S_8(35) = \{10, 35\}, \\
S_8(13) &= S_8(14) = S_8(22) = S_8(41) = \{13, 14, 22, 41\}, \\
S_8(15) &= S_8(30) = \{15, 30\}, \\
S_8(20) &= S_8(25) = \{20, 25\}, \\
S_8(21) &= S_8(33) = S_8(39) = S_8(42) = \{21, 33, 39, 42\}, \text{ and} \\
S_8(26) &= S_8(28) = S_8(37) = S_8(44) = \{26, 28, 37, 44\}.
\end{aligned}$$

We have the following observations.

1. For the 8-cyclotomic coset $S_8(9) = \{9, 18, 27, 36\}$, we have $S_8(9) = S_8(36) = S_8(-9)$ which implies that it is of type I. Moreover, it is of type II since $S_8(9) = S_8(27) = S_8(-18) = S_8(-2 \cdot 9)$. Subsequently, it is of type III since $S_8(9) = S_8(-36) = S_8(-2^2 \cdot 9)$.
2. For the 8-cyclotomic coset $S_8(5) = \{5, 40\}$, we have $S_8(5) = S_8(40) = S_8(-5)$ which implies that it is of type I. Since $S_8(5) \neq \{10, 35\} = S_8(35) = S_8(-10) = S_8(-2 \cdot 5)$, it is of type II'. Since $S_8(5) \neq \{20, 25\} = S_8(25) = S_8(-20) = S_8(-2^2 \cdot 5)$, it is of type III'.
3. For the 8-cyclotomic coset $S_8(3) = \{3, 6, 12, 24\}$, we have $S_8(3) \neq \{21, 33, 39, 42\} = S_8(42) = S_8(-3)$ which means that it is of type I'. Moreover, it is of type II' since $S_8(3) \neq \{21, 33, 39, 42\} = S_8(39) = S_8(-6) = S_8(-2 \cdot 3)$. Since $S_8(3) \neq \{21, 33, 39, 42\} = S_8(33) = S_8(-12) = S_8(-2^2 \cdot 3)$, it is of type III'.

Remark 2.12 ([6]). We have the following facts.

1. A q -cyclotomic coset of $\mathbb{Z}_m$ of type I always exists and its cardinality is either 1 or even.
2. For each $a \in \mathbb{Z}_m$, every element in $S_q(a)$ has the same order.
3. For each $a, b \in \mathbb{Z}_m$, if $\text{ord}(a) = \text{ord}(b)$, then $|S_q(a)| = \text{ord}_{\text{ord}(a)}(q) = \text{ord}_{\text{ord}(b)}(q) = |S_q(b)|$.

Definition 2.13. The Euler's totient function ϕ is a function defined by $\phi(n) = |\{k \mid 1 \leq k \leq n \text{ and } \gcd(n, k) = 1\}|$ for all positive integers n .

Theorem 2.14 ([6]). Let q be a prime power and let m be a positive integer such that $\gcd(q, m) = 1$. Then the number of q -cyclotomic cosets of $\mathbb{Z}_m$ is $\sum_{d|m} \frac{\phi(d)}{\text{ord}_d(q)}$, where ϕ is the Euler's totient function.

Example 2.15. From Theorem 2.7 and Lemma 2.8, we observe that

$$\mathbb{Z}_{45} = \bigcup_{d|45} A_d = A_1 \cup A_3 \cup A_5 \cup A_9 \cup A_{15} \cup A_{45},$$

where A_d denotes the set of elements in $\mathbb{Z}_m$ whose additive order is d . Precisely,

$$A_1 = \{0\}.$$

$$A_3 = \{15, 30\},$$

$$A_5 = \{9, 18, 27, 36\},$$

$$A_9 = \{5, 10, 20, 25, 35, 40\},$$

$$A_{15} = \{3, 6, 12, 21, 24, 33, 39, 42\}, \text{ and}$$

$$A_{45} = \{1, 2, 4, 7, 8, 11, 13, 14, 16, 17, 19, 22, 23, 26, 28, 29, 31, 32, 34, 37, 38, 41, 43, 44\}.$$

By observation from Remark 2.12(2), for each $d \in \{1, 3, 5, 9, 15, 45\}$, A_d can be partitioned into a union of 8-cyclotomic classes $S_8(a)$ as follows:

$$A_1 = S_8(0),$$

$$A_3 = S_8(15),$$

$$A_5 = S_8(9),$$

$$A_9 = S_8(5) \cup S_8(10) \cup S_8(20),$$

$$A_{15} = S_8(3) \cup S_8(21), \text{ and}$$

$$A_{45} = S_8(1) \cup S_8(2) \cup S_8(4) \cup S_8(7) \cup S_8(13) \cup S_8(26).$$

For each $a \in \mathbb{Z}_{45}$, the 8-cyclotomic coset $S_8(a)$ are described in Example 2.11. By Theorem 2.14, we have that the number of 8-cyclotomic cosets of $\mathbb{Z}_{45}$ is 14 which coincides with the discussion above.

Now we define the notation for the number of q -cyclotomic cosets of each type in $\mathbb{Z}_m$.

Definition 2.16. For each positive integer m and prime power q such that $\gcd(m, q) = 1$, let $t_I(m, q)$ and $t_{I'}(m, q)$ denote the number of q -cyclotomic cosets of type I and I' in $\mathbb{Z}_m$, respectively.

In addition, if $q = p^3$ is a prime-cubed integer, let $t_{II}(m, q)$, $t_{III}(m, q)$, $t_{II'}(m, q)$, and $t_{III'}(m, q)$ denote the number of q -cyclotomic cosets of type II, III, II', and III' in $\mathbb{Z}_m$, respectively.

Example 2.17. From Example 2.11, the following results can be easily derived.

1. $S_8(0), S_8(15) = S_8(30), S_8(9) = S_8(18) = S_8(27) = S_8(36)$ are of type I, II and III.
2. $S_8(5) = S_8(40), S_8(10) = S_8(35), S_8(20) = S_8(25)$ are of type I, II' and III'.
3. $S_8(1) = S_8(8) = S_8(17) = S_8(19), S_8(2) = S_8(16) = S_8(34) = S_8(38), S_8(3) = S_8(6) = S_8(12) = S_8(24), S_8(4) = S_8(23) = S_8(31) = S_8(32), S_8(7) = S_8(11) = S_8(29) = S_8(43), S_8(13) = S_8(14) = S_8(22) = S_8(41), S_8(21) = S_8(33) = S_8(39) = S_8(42), S_8(26) = S_8(28) = S_8(37) = S_8(44)$ are of type I', II' and III'.

Therefore, we have $t_I(45, 8) = 6$, $t_{II}(45, 8) = t_{III}(45, 8) = 3$ and $t_{I'}(45, 8) = 8$, $t_{II'}(45, 8) = t_{III'}(45, 8) = 11$.

Remark 2.18. From the definitions and discussion above, we have

$$\begin{aligned} \sum_{d|m} \frac{\phi(d)}{\text{ord}_d(q)} &= t_I(m, q) + t_{I'}(m, q) \\ &= t_{II}(m, q) + t_{II'}(m, q) \\ &= t_{III}(m, q) + t_{III'}(m, q). \end{aligned}$$

Definition 2.19. For each positive integer q , let $\chi_q : \mathbb{N} \rightarrow \{0, 1\}$ be a function defined by

$$\chi_q(j) = \begin{cases} 0 & \text{if there exists an integer } s \geq 1 \text{ such that } j \mid q^s + 1, \\ 1 & \text{otherwise.} \end{cases}$$

Example 2.20. Let $p = 2$. Then we have the following values for $\chi_2(t)$.

1. Since $2^{3 \cdot 2} + 1 = 65$ and $5 \mid 65$, we have $\chi_2(5) = 0$.
2. Since $2^{3s} + 1 \equiv 1^s + 1 \equiv 2 \pmod{7}$, then $7 \nmid 2^{3s} + 1$ for all $s \geq 0$. Hence, $\chi_2(7) = 1$.

Lemma 2.21 ([3]). Let q be a prime power and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. Then $S_q(a)$ is of type I' if and only if $\chi_q(\text{ord}(a)) = 1$

Theorem 2.22 ([3]). For a positive integer m and a prime power q such that $\gcd(m, q) = 1$, we have

$$t_{I'}(m, q) = \sum_{d \mid m} \chi_q(d) \frac{\phi(d)}{\text{ord}_d(q)}.$$

Chapter 3

Prime-Cubed Cyclotomic Cosets in $\mathbb{Z}_m$

In this chapter, characterization and enumeration of q -cyclotomic cosets of $\mathbb{Z}_m$ of types II, III, II', and III' are discussed in the case where q is a prime-cubed integer.

3.1 Types of Prime-Cubed Cyclotomic Cosets

Let p be a prime and let $q = p^3$ be a prime-cubed integer. Let m be a positive integer such that $\gcd(p, m) = 1$. In this section, characterization of q -cyclotomic cosets in $\mathbb{Z}_m$ of types II, III, II', and III' are discussed as well as relations among them and q -cyclotomic cosets of types I and I'.

In $\mathbb{Z}_m$, the q -cyclotomic coset $S_q(0) = \{0\}$ is of type I, II, and III. Hence, we have the following result.

Proposition 3.1. Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Then a q -cyclotomic coset of $\mathbb{Z}_m$ of type I, II and III always exist.

A relation between q -cyclotomic cosets in $\mathbb{Z}_m$ of types II and III is determined in the next theorem.

Theorem 3.2. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then $S_q(a)$ is of type II if and only if $S_q(a)$ is of type III .

Proof. Suppose that $S_q(a)$ is of type II . Precisely, we have $S_q(a) = S_q(-pa)$. Then $-pa = q^i a$ for some $0 \leq i < \text{ord}_{\text{ord}(a)}(q)$ which implies $qa = -p^2 q^i a$. Therefore, $S_q(a) = S_q(qa) = S_q(-p^2 a)$ and hence $S_q(a)$ is of type III .

Conversely, suppose that $S_q(a)$ is of type III . Then $S_q(a) = S_q(-p^2 a)$, i.e., $-p^2 a = q^i a$ for some $0 \leq i < \text{ord}_{\text{ord}(a)}(q)$. This implies that $qa = -pq^i a$. Therefore, $S_q(a) = S_q(qa) = S_q(-pa)$ is of type II as desired. $\square$

Corollary 3.3. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then $S_q(a)$ is of type II' if and only if $S_q(a)$ is of type III' .

Lemma 3.4. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. If $S_q(a)$ is of type II or III , then it is of type I .

Proof. Suppose that $S_q(a)$ is of type II or III . By Theorem 3.2, it suffices to prove for the case where $S_q(a)$ is of type II . Then $S_q(a) = S_q(-pa) = S_q(-p^2 a)$ which implies that $-pa = q^i(-p^2 a)$ for some $0 \leq i < \text{ord}_{\text{ord}(a)}(q)$, i.e., $-a = q^i(-pa)$. Therefore, $S_q(-a) = S_q(-pa) = S_q(a)$ is of type I . $\square$

We note that the converse of the above lemma does not need to be true.

Example 3.5. For the 8-cyclotomic coset $S_8(5) = \{5, 40\}$ in $\mathbb{Z}_{45}$, we have $S_8(5) = S_8(40) = S_8(-5)$ which implies that it is of type I . While

$$S_8(5) \neq \{10, 35\} = S_8(35) = S_8(-10) = S_8(-2 \cdot 5)$$

is of type II' and

$$S_8(5) \neq \{20, 25\} = S_8(25) = S_8(-20) = S_8(-2^2 \cdot 5)$$

is of type III' .

Corollary 3.6. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. If $S_q(a)$ is of type I' , it is of type II' and III' .

Remark 3.7. From Theorem 3.2 and Corollary 3.6. It is easily seen that $t_I(m, q) \geq t_{II}(m, q) = t_{III}(m, q)$ and $t_{I'}(m, q) \leq t_{II'}(m, q) = t_{III'}(m, q)$ for all prime-cubed integers q and positive integers m such that $\gcd(m, q) = 1$.

Corollary 3.8. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. The cardinality of q -cyclotomic coset in $\mathbb{Z}_m$ of type I, II or III is either 1 or even.

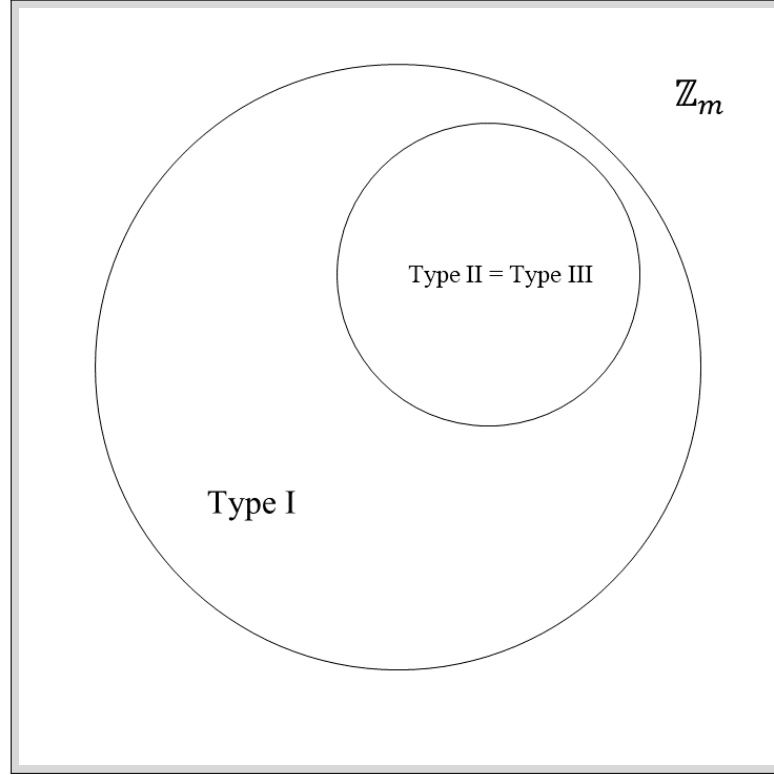
Proof. Assume that a q -cyclotomic coset in $\mathbb{Z}_m$ is of type I, II or III. Then by Lemma 3.4, it is of type I. Hence, the result follows from Remark 2.12(1). $\square$

Corollary 3.9. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. If the cardinality of a q -cyclotomic coset in $\mathbb{Z}_m$ is odd and greater than 1, then it is of type I' , II' and III' .

We note that the converse of the above corollaries do not need to be true.

Example 3.10. For the 27-cyclotomic coset $S_{27}(2) = \{2, 6\}$ in $\mathbb{Z}_{16}$, we have $|S_{27}(2)| = 2$ but it is of type I' , II' and III' .

Relations among q -cyclotomic cosets of types I, II and III are summarized in the following Venn diagram. Note that q -cyclotomic cosets of I' , II' and III' are complement of types I, II and III, respectively.



The characterization of q -cyclotomic cosets of type II (and type III) is given as follows.

Lemma 3.11. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then the q -cyclotomic coset $S_q(a)$ is of type II if and only if $\text{ord}(a) \mid p^{3s+1} + 1$ for some integer $s \geq 0$.

Proof. Suppose that $S_q(a)$ is of type II. Then $S_q(a) = S_q(-pa)$. Equivalently, $a \equiv q^s(-pa) \pmod{m}$ for some integers $s \geq 0$. Therefore, $0 \equiv (q^s p + 1)a \equiv (p^{3s+1} + 1)a \pmod{m}$ which is equivalent to $0 \equiv q^s p + 1 \equiv p^{3s+1} + 1 \pmod{\text{ord}(a)}$ for some integers $s \geq 0$. Hence, the result follows.

Conversely, suppose that $\text{ord}(a) \mid p^{3s+1} + 1$ for some integer $s \geq 0$. Equivalent, $p^{3s+1} \equiv -1 \pmod{\text{ord}(a)}$. Multiplying $p^2 a$, we have $q^{s+1} a \equiv -p^2 a \pmod{a \text{ord}(a)}$. Since $a \text{ord}(a)$ is multiple of m , we have that $S_q(a) = S_q(-p^2 a)$. Therefore, $S_q(a)$ is of type III and hence it is of type II by Theorem 3.2. $\square$

Theorem 3.12. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a, b \in \mathbb{Z}_m$. If $\text{ord}(a) = \text{ord}(b)$, then the q -cyclotomic cosets $S_q(a)$ and $S_q(b)$ are of the same type.

Proof. Assume that $\text{ord}(a) = \text{ord}(b)$.

First, assume that $S_q(a)$ is of type I. From Lemma 2.21, there exists $s \geq 1$ such that $\text{ord}(a) \mid q^s + 1$. Since $\text{ord}(a) = \text{ord}(b)$, we have $\text{ord}(b) \mid q^s + 1$ which is equivalent to $b \cdot \text{ord}(b) \mid b \cdot (q^s + 1)$. Since $m \mid b \cdot \text{ord}(b)$, it follows that $m \mid b \cdot (q^s + 1)$ which is $q^s b \equiv -b \pmod{m}$. Therefore, $S_q(b) = S_q(-b)$ is of type I.

Next, suppose that $S_q(a)$ is of type II (resp, type III by Theorem 3.2). From Lemma 3.11, there exists an integer $s \geq 0$ such that $\text{ord}(a) \mid pq^s + 1$. Since $\text{ord}(a) = \text{ord}(b)$, it follows that $\text{ord}(b) \mid pq^s + 1$ which is equivalent to $b \cdot \text{ord}(b) \mid b \cdot (pq^s + 1)$. Since $m \mid b \cdot \text{ord}(b)$, it follows that $m \mid b \cdot (pq^s + 1)$ which is $q^{s+1}b \equiv -p^2b \pmod{m}$. Therefore $S_q(b) = S_q(-p^2b)$, hence it is of type III and also of type II. $\square$

3.2 Enumeration of Prime-Cubed Cyclotomic Cosets

The enumeration of q -cyclotomic cosets of each type will be discussed in this section. Some number theoretical results can be obtained as a by-product.

We begin with the following useful lemma.

Lemma 3.13. Let a be a positive integer and p be a prime number. The following statements are equivalent:

- (1) $a \mid p^{3s-2} + 1$ for some integer $s \geq 0$.
- (2) $a \mid p^{3s-1} + 1$ for some integer $s \geq 0$.
- (3) $a \mid p^{3s+1} + 1$ for some integer $s \geq 0$.
- (4) $a \mid p^{3s+2} + 1$ for some integer $s \geq 0$.

Proof. (1) $\rightarrow$ (2). Suppose $a \mid p^{3s-2} + 1$ for some $s \geq 0$. Since $\gcd(a, p) = 1$, we have $p^{3s-2} + 1 \equiv 0 \pmod{a}$ which is equivalent to $p^{3s} \equiv -p^2 \pmod{a}$. It follows that $S_{p^3}(1) = S_{p^3}(-p^2)$ in $\mathbb{Z}_a$. Hence, $S_{p^3}(1)$ is of type III. By Theorem 3.2, $S_{p^3}(1)$ is of type II which implies that $S_q(1) = S_q(-p)$. Hence, $p^{3t} \equiv -p \pmod{a}$ for some $t \geq 0$ which implies that $a \mid p^{3t-1} + 1$.

(2) $\rightarrow$ (3). Suppose $a \mid p^{3s-1} + 1$ for some $s \geq 0$. Since $\gcd(a, p) = 1$, we have $p^{3s-1} + 1 \equiv 0 \pmod{a}$ which is equivalent to $p^{3s} \equiv -p \pmod{a}$. It follows that

$S_{p^3}(1) = S_{p^3}(-p)$ in $\mathbb{Z}_a$. Therefore, $1 \equiv p^{3t}(-p) \pmod{a}$ for some $t \geq 0$ which implies that $a \mid p^{3t+1} + 1$.

(3) $\rightarrow$ (4). Suppose $a \mid p^{3s+1} + 1$ for some $s \geq 0$. Since $\gcd(a, p) = 1$, we have $p^{3s+1} + 1 \equiv 0 \pmod{a}$ which is equivalent to $p^{3(s+1)} \equiv -p^2 \pmod{a}$. It follows that $S_{p^3}(1) = S_{p^3}(-p^2)$ in $\mathbb{Z}_a$. Therefore, $1 \equiv p^{3t}(-p^2) \pmod{a}$ for some $t \geq 0$ which implies that $a \mid p^{3t+2} + 1$.

(4) $\rightarrow$ (1). Suppose $a \mid p^{3s+2} + 1$ for some $s \geq 0$. Since $\gcd(a, p) = 1$, we have $p^{3s+2} + 1 \equiv 0 \pmod{a}$ which is equivalent to $p^{3(s+1)} \equiv -p \pmod{a}$. It follows that $S_{p^3}(1) = S_{p^3}(-p)$ in $\mathbb{Z}_a$. Therefore, $1 \equiv p^{3(t-1)}(-p) \pmod{a}$ for some $t \geq 1$ which implies that $a \mid p^{3t-2} + 1$.

□

Based on the above characterization, we define the following characteristic function.

Definition 3.14. For each prime p , let $\lambda_p : \mathbb{N} \rightarrow \{0, 1\}$ be a function defined by

$$\lambda_p(t) = \begin{cases} 0 & \text{if there exists an integer } s \geq 0 \text{ such that } t \mid p^{3s+1} + 1, \\ 1 & \text{otherwise.} \end{cases} \quad (\star)$$

Example 3.15. Let $p = 2$. Then we have the following values for $\lambda_2(t)$.

1. Since $2^{3 \cdot 3+1} + 1 = 1025$ and $5 \mid 1025$, we have $\lambda_2(5) = 0$.
2. We show that $9 \nmid 2^{3s+1} + 1$ for all integers $s \geq 0$. If s is odd then

$$2^{3s+1} + 1 \equiv 2^{3(2n+1)+1} + 1 \equiv 2^{6n+4} + 1 \equiv 2^{6n} \cdot 2^4 + 1 \equiv 1^n \cdot 7 + 1 \equiv 8 \pmod{9}$$

for some $n \in \mathbb{Z}$. In the other hand, if s is even then

$$2^{3s+1} + 1 \equiv 2^{3(2n)+1} + 1 \equiv 2^{6n+1} + 1 \equiv 2^{6n} \cdot 2 + 1 \equiv 1^n \cdot 2 + 1 \equiv 3 \pmod{9}$$

for some $n \in \mathbb{Z}$. Therefore $9 \nmid 2^{3s+1} + 1$ for all $s \geq 0$. Hence, $\lambda_2(9) = 1$.

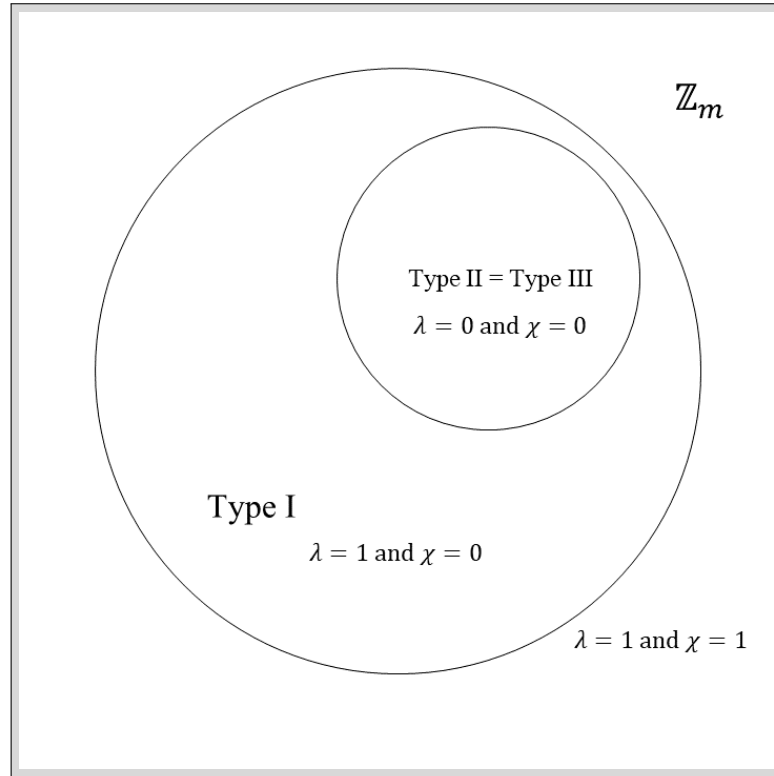
From Lemma 3.11 and Definition 3.14, we have the following result.

Theorem 3.16. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. Then the q -cyclotomic coset of $S_q(a)$ is of type II (and also of type III) if and only if $\lambda_p(\text{ord}(a)) = 0$.

Proof. Suppose that $S_q(a)$ is of type II. We have that it is also of type III. Then $S_q(a) = S_q(-pa) = S_q(-p^2a)$. Which is equivalent to $a \equiv q^i(-pa) \equiv q^j(-p^2a) \pmod{m}$. Therefore $0 \equiv q^i p + 1 \equiv q^j p^2 + 1 \pmod{\text{ord}(a)}$. Which completes the proof by Lemma 3.13.

Conversely, suppose that $\lambda_p(\text{ord}(a)) = 0$. Then $\text{ord}(a) \mid p^{3s+1} + 1$ for some $s \geq 1$. Which is equivalent to $p^{3s+1} \equiv -1 \pmod{\text{ord}(a)}$. By multiplying p^2a , we have $q^{s+1}a \equiv -p^2a \pmod{a \cdot \text{ord}(a)}$. Since $a \cdot \text{ord}(a)$ is multiple of m , thus $S_q(a) = S_q(-p^2a)$ of $\mathbb{Z}_m$. Therefore $S_q(a)$ is of type III, hence it is of type II. $\square$

Relations of λ and χ among q -cyclotomic cosets of types I, II and III are summarized in the following Venn diagram. Note that q -cyclotomic cosets of I', II' and III' are complement of types I, II and III, respectively.



Example 3.17. Consider the cyclic group $a \in \mathbb{Z}_{45}$ in Example 2.11 and Example 2.17. For each $d \in \{1, 3, 5, 9, 15, 45\}$, the value of $\lambda_2(d)$ can be computed as follows.

$S_8(0)$ is of type II and $\text{ord}(0) = 1$, hence $\lambda_2(1) = 0$.

$S_8(15)$ is of type II and $\text{ord}(15) = 3$, hence $\lambda_2(3) = 0$.

$S_8(9)$ is of type II and $\text{ord}(9) = 5$, hence $\lambda_2(5) = 0$.

$S_8(5)$ is of type II' and $\text{ord}(5) = 9$, hence $\lambda_2(9) = 1$.

$S_8(3)$ is of type II' and $\text{ord}(3) = 15$, hence $\lambda_2(15) = 1$.

$S_8(1)$ is of type II' and $\text{ord}(1) = 45$, hence $\lambda_2(45) = 1$.

Thus the number of q -cyclotomic cosets of type II and III can be determined as follows.

Theorem 3.18. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Then the number of q -cyclotomic cosets of type II' and type III' is

$$t_{II'}(m, q) = t_{III'}(m, q) = \sum_{d|m} \lambda_p(d) \frac{\phi(d)}{\text{ord}_d(q)}.$$

Proof. Since the cosets of the same order are of the same type, we consider the cosets whose an additive order is $d \mid m$. Hence we have $\frac{\phi(d)}{\text{ord}_d(q)}$ cosets. If they are of type II' and III', $\lambda_p(d) = 1$. Otherwise, $\lambda_p(d) = 0$. Hence the number of type II' and III' cosets having an additive order d is $\lambda_p(d) \frac{\phi(d)}{\text{ord}_d(q)}$. Using a summation, the number of all cosets follows. $\square$

Example 3.19. From Example 2.15, for each $d \in \{1, 3, 5, 9, 15, 45\}$, A_d denotes the set of elements having an additive order d in $\mathbb{Z}_m$. The set A_d can be partitioned by $S_8(a)$ as follows:

$$A_1 = S_8(0),$$

$$A_3 = S_8(15),$$

$$A_5 = S_8(9),$$

$$A_9 = S_8(5) \cup S_8(10) \cup S_8(20),$$

$A_{15} = S_8(3) \cup S_8(21)$ and

$A_{45} = S_8(1) \cup S_8(2) \cup S_8(4) \cup S_8(7) \cup S_8(13) \cup S_8(26)$.

Together with Theorem 3.18, we have

$$t_{II'}(45, 8) = \sum_{d|45} \lambda_p(d) \frac{\phi(d)}{\text{ord}_d(8)} = 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 3 + 1 \cdot 2 + 1 \cdot 6 = 11.$$

Lemma 3.20. Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. Then $\lambda_p(\text{ord}(a)) = 0$ if and only if the q -cyclotomic coset $S_q(a)$ is of type I, II and III.

Proof. Suppose that $\lambda_p(\text{ord}(a)) = 0$. By Theorem 3.16, we have $S_q(a)$ is of type II and III. Thus by Lemma 3.4, it is of type I. Conversely, suppose that the q -cyclotomic coset $S_q(a)$ is of type I, II and III then $\lambda_p(\text{ord}(a)) = 0$ by Theorem 3.16. $\square$

From Lemma 3.20, we get the following result in number theory.

Corollary 3.21. Let p be a prime number and a be a positive integer such that $\gcd(a, p) = 1$. If $a \mid p^{3s+1} + 1$ for some $s \geq 0$, then for each $r \in \mathbb{Z}$ there exists $k \in \mathbb{Z}$ such that $a \mid p^{3k+r} + 1$.

Proof. Let a be a positive integer, considering in $\mathbb{Z}_{a^2}$, $\text{ord}(a) = a$. Suppose that $a \mid p^{3s+1} + 1$. Then $0 = \lambda_p(a) = \lambda_p(\text{ord}(a))$, $S_q(a)$ is of type I, II and III. Thus $\chi_{p^3}(\text{ord}(a)) = \chi_{p^3}(a) = 0$, that is, there exists $s \geq 1$ such that $a \mid p^{3s} + 1$. Applying Lemma 3.13, there exist $s_0, s_1, s_2 \geq 1$ satisfying $a \mid p^{3s_0} + 1$, $a \mid p^{3s_1+1} + 1$ and $a \mid p^{3s_2+2} + 1$. Let $r \in \mathbb{Z}$, by division algorithm, there exist $c, d \in \mathbb{Z}$ such that $r = 3c + d$ for $0 \leq d < 3$. We consider d in 3 cases.

Case $d = 0$, choose $k = s_0 - c$ thus $p^{3k+r} + 1 = p^{3s_0-3c+3c+0} + 1 = p^{3s_0} + 1$.

Case $d = 1$, choose $k = s_1 - c$ thus $p^{3k+r} + 1 = p^{3s_1-3c+3c+1} + 1 = p^{3s_1+1} + 1$.

Case $d = 2$, choose $k = s_2 - c$ thus $p^{3k+r} + 1 = p^{3s_2-3c+3c+2} + 1 = p^{3s_2+2} + 1$.

Therefore $a \mid p^{3k+r} + 1$ for all $r \in \mathbb{Z}$. $\square$

Chapter 4

Conclusion and Remarks

In this chapter, the main results are summarized. Relations between q -cyclotomic cosets of type II and III are shown.

Theorem 3.2 Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then $S_q(a)$ is of type *II* if and only if $S_q(a)$ is of type *III*.

Some relations between q -cyclotomic cosets of types I, II and III are shown.

Lemma 3.4 Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$. If $S_q(a)$ is of type *II* or *III*, then it is of type *I*.

Characterizations of q -cyclotomic cosets of type II and also type III are shown.

Lemma 3.11 Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Let $a \in \mathbb{Z}_m$, then the q -cyclotomic coset $S_q(a)$ is of type II if and only if $\text{ord}(a) \mid p^{3s+1} + 1$ for some integer $s \geq 0$.

Enumerations of q -cyclotomic cosets of type II' and type III' are shown in the following theorem. Note that the enumerations of q -cyclotomic cosets of type II and type III can be derived directly from their relations.

Theorem 3.18 Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Then the number of q -cyclotomic cosets of type

II' and type III' is

$$t_{II'}(m, q) = t_{III'}(m, q) = \sum_{d|m} \lambda_p(d) \frac{\phi(d)}{\text{ord}_d(q)},$$

where $\lambda_p(d)$ is defined in $(\star)$.

Lastly, one of the number theoretical results are shown.

Lemma 3.13 Let a be a positive integer and p be a prime number. The following statements are equivalent:

- (1) $a \mid p^{3s-2} + 1$ for some integer $s \geq 0$.
- (2) $a \mid p^{3s-1} + 1$ for some integer $s \geq 0$.
- (3) $a \mid p^{3s+1} + 1$ for some integer $s \geq 0$.
- (4) $a \mid p^{3s+2} + 1$ for some integer $s \geq 0$.

In general, the study of different types of q -cyclotomic cosets in $\mathbb{Z}_m$ where $q = p^m$, p is a prime and m is a positive integer is interesting.

Reference

- [1] Y. Jia, S. Ling, C. Xing, On self-dual cyclic codes over finite fields, *IEEE Trans. Inf. Theory*, **57** (2011) 2243–2251.
- [2] S. Jitman., S. Ling, Quasi-abelian codes, *Designs, Codes and Cryptography*, **74** (2015) 511–531.
- [3] S. Jitman., S. Ling, H.-W. Liu, X. Xie, Abelian codes in principal ideal group algebras, *IEEE Trans. Inf. Theory*, **59** (2013) 3046–3058.
- [4] S. Jitman., S. Ling, P. Solé, Hermitian self-dual abelian codes, *IEEE Trans. Inf. Theory*, **60** (2014) 1496–1507.
- [5] W. K. Nicholson, *Introduction to Abstract Algebra*, A John Wiley & Sons, Inc., publication, 4th edition, 2012.
- [6] S. Prugsapitak., S. Jitman., Enumeration of self-dual cyclic codes of some specific lengths over finite fields, *Discrete Mathematics, Algorithms and Applications*, **10** (2018) Article ID: 1850031.

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